

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

B.Tech. (CL/EC/ME) Sem - I Examination, 2010-11 (CE/IT/EE)

ME - 101 Engineering Graphics

Date: 18.12.2010, Saturday

Time: 10:30 a.m. to 12:30 p.m.

Maximum Marks: 70

Instructions: (i) Attempt **all** the questions.

(ii) Figures to the right indicate **full** marks.

Section 1

- 1.(a) Construct a scale of 1.5 inches=1 foot to show inches and long enough to measure up to 4 feet. [04]
Show 2'-10" (2 feet 10 inches) on the same scale.

- 1.(b) A circle of 25 mm radius rolls on the circumference of another circle of 150 mm diameter and outside it. Draw the locus of the point P on the circumference of the rolling circle for one complete revolution of it. Name the curve & draw tangent and normal to the curve. [06]

2. The end A of line AB is in the HP and 25 mm behind VP. The end B is in the VP and 50 mm above HP. The distance between ends projectors is 75 mm. Draw the projections of line AB; determine its true length and inclinations with HP and VP. [10]

OR

2. The end projectors of line PQ are 22 mm apart. P is 12 mm in front of VP and 12 mm above HP. The point Q is 6 mm in front of VP and 40 mm above HP. Draw its projections. Find its true length and inclinations with the HP and VP. [10]

3. Fig. 1 shows pictorial view of an object. Draw (1) Front view in the direction 'X', (2) Right hand side view and (3) Top view [15]

OR

3. Fig. 2 shows pictorial view of an object. By Using first angle projection method, Draw (1) Half Sectional Front view in the direction 'X', (2) Top view. [15]

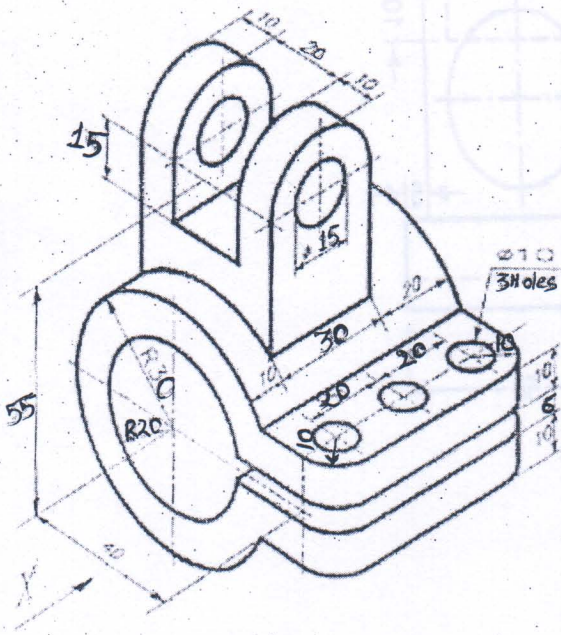


Fig .1

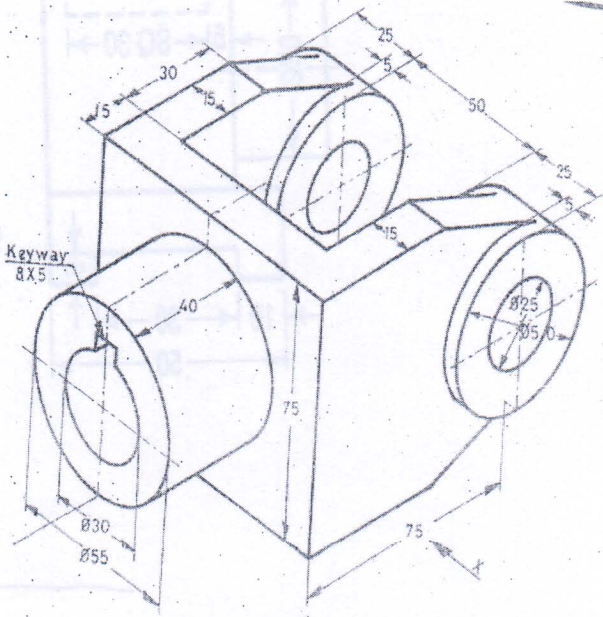


Fig.2

Section 2

- 4(a) Draw the projection of a regular hexagon of 25 mm side, having one of its sides in the H.P. and inclined at 60° to the V.P. and its surface making an angle of 45° with the H.P. [07]

- 4(b) A pentagonal pyramid, base 25 mm side and axis 50 mm long has one of its triangular faces in the V.P. and the edge of the base contained by that face makes an angle of 30° with the H.P. Draw its Projections. [08]

OR

- 4(a) A circular plate of negligible thickness and 50 mm diameter appears as an ellipse in the front view, having its major axis 50 mm long and minor axis 30 mm long. Draw its top view when the major axis of the ellipse is horizontal. [07]

- 4(b) Draw the projection of a cube of 30 mm long edges resting on the ground on one of its corners with a solid diagonal perpendicular to the V.P and parallel to V.P. [08]

- 5 Draw the projections of a cone resting on the ground on its base and show on them, the shortest path by which a point P, starting from a point on the circumference of the base and moving around the cone will return to the same point. Base of cone 65 mm diameter ; axis 75 mm long. [10]

OR

- 5 A hexagonal prism has a face on the ground and the axis parallel to the V.P. It is cut by a vertical section plane, which makes an angle of 45° with xy and which cuts the axis at a point 20 mm from one of its ends. Draw its sectional front view and the true shape of the section. Side of base 25 mm long; height 65mm. [10]

- 6 Draw the isometric view of object shown in Fig 3 [10]

